

Comparative Study of Brain Tumor Types Interaction with Various Edge Detection Operators

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Abstract- Brain tumors, whether malignant or benign, pose significant challenge in the medical field due to their often-elusive nature and the severe implications they carry. Diagnosing these tumors accurately requires the expertise of doctor's specialists, who rely on a combination of advanced imaging techniques and clinical evaluations. Machine learning has emerged as a powerful ally in this battle, particularly in the realm of edge detection and image analysis. Various edge detection operators, such as the Sobel, Canny, and Kirsch operators, play a crucial role in enhancing the clarity of MRI scans, making it easier to delineate the tumor boundaries. This paper conducts a comparative analysis of various operators to determine their efficacy in detecting brain tumors, thereby highlighting the potential of machine learning to significantly improve diagnostic accuracy and treatment outcomes.

Keywords—Brain Tumor, Diagnosis, Machine learning, MRI scans, edge detection operators, Accuracy, Outcomes

I. INTRODUCTION

Brain tumors are complex and often fatal, requiring precise diagnostics. Machine learning, especially edge detection operators like Sobel, Canny, and Kirsch, enhances MRI analysis by better outlining tumors. Pradeep Kumar Mallick, et al. [1], comparison between various operators were done for edge detection of a tumor/cyst in brain, following which the research terminates with distance transform calculation, and claims that Canny operators performs best than all others. SangeethaG, et al. [2], provides insights to edge detection techniques to extract regions of the brain from MRI scans. Sobel and Prewitt operators are used to detect edges. Further morphological operations are used to make up for the erosion and dilation of unwanted parts. D. Haritha, compares between various brain tumor detection techniques alternately combining morphological operations, K-means algorithm, Local Binary Pattern, edge detection and more, simultaneously. The paper also gives the time measure taken by each of these techniques. [3] Warqaa Shaher AlAzawee, et al., taken Lenna's image and a noisy fingerprint image, they are seen to apply the concept of erosion, wherein, results are

compared across operators like, Sobel, Prewitt, Canny and Cellular Automata. It's seen that disk-shaped morphological operators are used to identify fine edges in binary images produced through Otsu's thresholding method. Remya Ajai A S, et al, [2022], taken Kaggle's Brain Tumor Dataset, its proven that 8-Sobel detection was the most suitable for tumor detection in brain, then the original Sobel, Prewitt, Roberts, Canny and Laplacian of Gaussian (LoG) operators. K.Srisabarimani, et al., [8], discusses about edge detection, noise reduction and feature extraction and other soft computing techniques used in traditional edge detection mechanisms.

II. DATASET

Brain Tumor Classification (MRI), by Sartaj, from Kaggle is utilised for this experiment. The version 2 holds a data of 93.08 MB with about 3264 files. The images are already split into Training and Testing folders. Each folder has four sub-folders, viz, Pituitary, Glioma, Meningioma and No Tumor [Fig.1.].

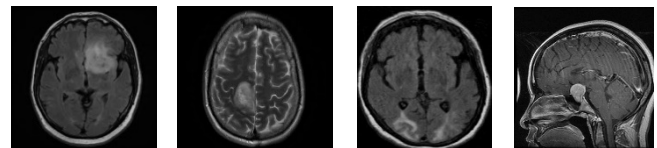


Fig.1. Types of Sub – classes (a) Glioma (b) Meningioma (c) No Tumor (d) Pituitary

III. EDGE DETECTION

One among the three primary types of grey level discontinuities, is edges. Edges are crucial in image processing. They provide the geometric boundary of the object/image. An edge consists of a sequence of connected pixels that form the outline between two areas with differing grey values. These pixels, known as edge points, delineate the transition between regions. For an edge to be defined effectively, it's essential to measure grey level changes accurately. Most edges are unique in space, i.e., their position and orientation remain the same in space when viewed from different points. Fig.2. illustrates the process involved in edge detection. When an edge is detected, the unnecessary details

are removed, while only the important structural information is retained. In, short an edge is a local concept which represents only significant intensity transitions. Table 1. gives the types of detectors used in this paper.

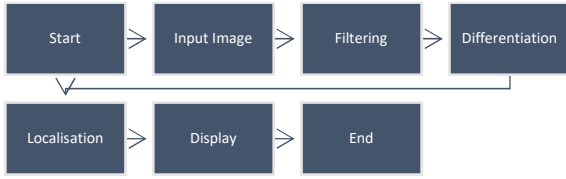


Fig.2. Edge Detection Process
Table 1. Types of Edge Detectors

ORDER OF DERIVATION	TYPES OF EDGE DETECTORS	OPERATORS
First	Derivative Types	Sobel
		Roberts
		Scharr
	Template Matching	Kirsch masks Frie-Chen masks
Second	Gaussian Derivatives	Canny
		Laplacian of Gaussian (LoG)

Scharr

This is a filtering method used to identify and highlight gradient features and performs quite similar to the Sobel filter. The above mentioned kernels are used for detection. The Scharr filter exhibits less rotational variance compared to the Sobel filter, which in turn outperforms the Prewitt filter in this regard.

Kirsch Masks

Kirsch masks are called compass masks because they're obtained by taking one mask and rotating it to eight major directions: north, north west, west, south west, south, south east, east and north east.

Frei – Chen Masks

Any image can be considered as the weighted sum of the nine Frei – Chen masks. The weights are obtained by a process called projecting process by overlapping a 3x3 image onto each mask by summing the multiplication of coincident terms. The first four masks represent the edge space, the next four represent the line subspace, and the last one represents the average subspace.

Canny

The Canny operator is an advanced edge detection technique used in image processing to identify the boundaries of objects within an image. It was developed by John F. Canny in 1986 and is known for its effectiveness and robustness.

Noise Reduction:

The input image is initially smoothed with a Gaussian filter, to irradiate unnecessary textures.

Gradient Calculation:

Now exposing the clear image to a Sobel operator, gradient intensity and direction will be resulted, thus helping in identifying edges.

Non-Maximum Suppression:

Refines the edges by conserving only the local maxima in the gradient image and thins them to their precise locations.

Double Thresholding:

Two threshold, high and low, are applied to categorize edges into strong, weak and irrelevant. Strong edges and weaker edges linked to strong edges are retained, and irrelevant edges are discarded.

Edge Tracking by Hysteresis:

The final step ensures edge continuity by connecting weak edges to strong ones and discarding isolated weak edges.

Laplacian of Gaussian (Marr – Hildrith Operator) – LoG

To reduce the noise sensitivity of the Laplacian operator, the LoG operator is commonly used. The process begins with blurring the image using a Gaussian filter, followed by applying the Laplacian. The Gaussian function helps in reducing noise, which in turn minimizes the detection of false edges when the Laplacian is applied.

IV. FLOW OF WORK

The figure, Fig.3. below directs on how to work to achieve the aim.

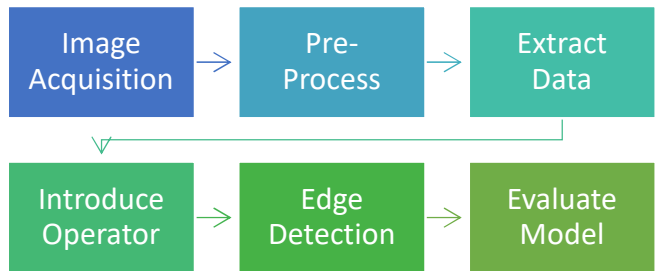


Fig.3. Flowchart to analyse the reaction of brain tumor images with various morphological operator

V. EXPERIMENTAL OUTCOMES

The dataset was exposed to the mentioned operators, not simultaneously. And the edge detection was carried out separately on both the axes, then was cumulated. Also, here comes the performance metrics for each of the experiment.

Sobel Operator

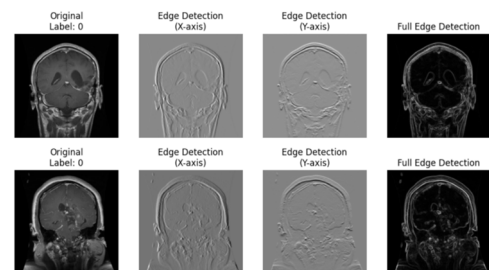


Fig.A1. Edge detection using Sobel Operator

Accuracy: 0.68
Precision: 0.81
Recall: 0.68
F1-Score: 0.63
Matthews Correlation Coefficient: 0.6108
Cohen's Kappa: 0.5565

	precision	recall	f1-score	support
glioma_tumor	1.00	0.18	0.31	100
meningioma_tumor	0.81	0.95	0.88	115
no_tumor	0.51	1.00	0.67	105
pituitary_tumor	0.97	0.46	0.62	74
accuracy			0.68	394
macro avg	0.82	0.65	0.62	394
weighted avg	0.81	0.68	0.63	394

Fig.A2. Evaluated Values of the Model – I

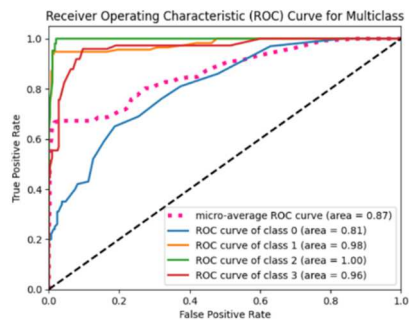


Fig.A3. AUC and ROC Curves (I)

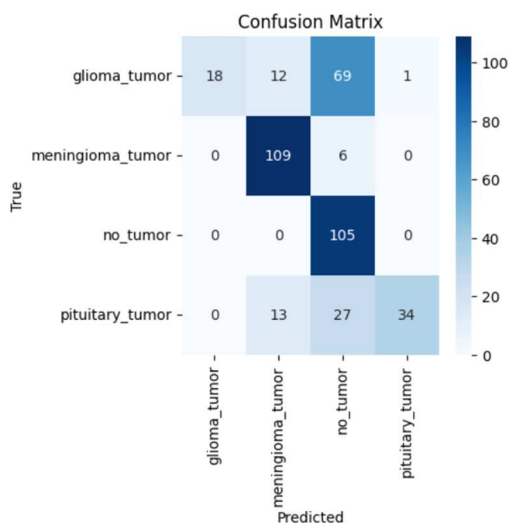


Fig.A4. Confusion Matrix (I)

Accuracy: 0.70
Precision: 0.79
Recall: 0.70
F1-Score: 0.66
Matthews Correlation Coefficient: 0.6345
Cohen's Kappa: 0.5921

	precision	recall	f1-score	support
glioma_tumor	0.95	0.21	0.34	100
meningioma_tumor	0.80	0.97	0.88	115
no_tumor	0.56	1.00	0.71	105
pituitary_tumor	0.87	0.53	0.66	74
accuracy			0.70	394
macro avg	0.80	0.68	0.65	394
weighted avg	0.79	0.70	0.66	394

Fig.B2. Evaluated Values of the Model – II

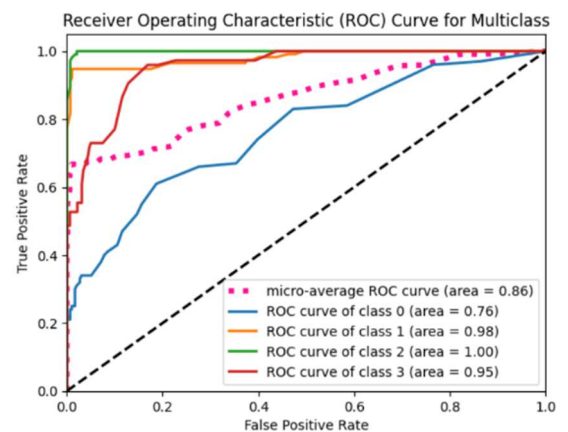


Fig.B3. AUC and ROC Curves (II)

Roberts Zero - Crossing

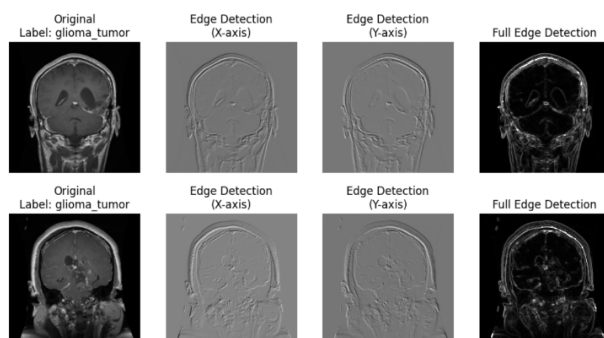


Fig.B1. Edge Detection using Roberts Zero – Crossing Operator

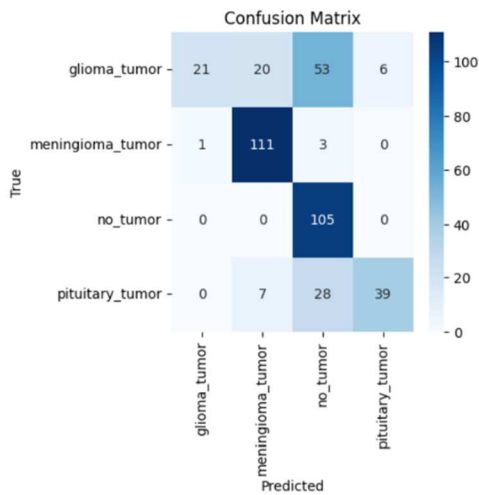
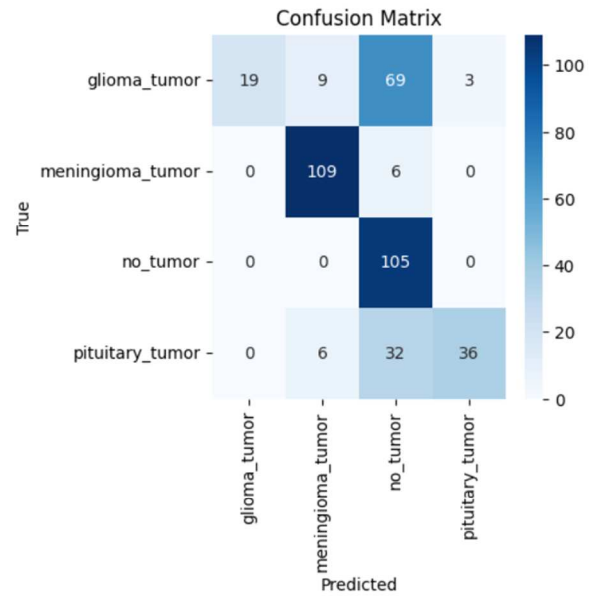


Fig.B4. Confusion Matrix (II)



Scharr Operator

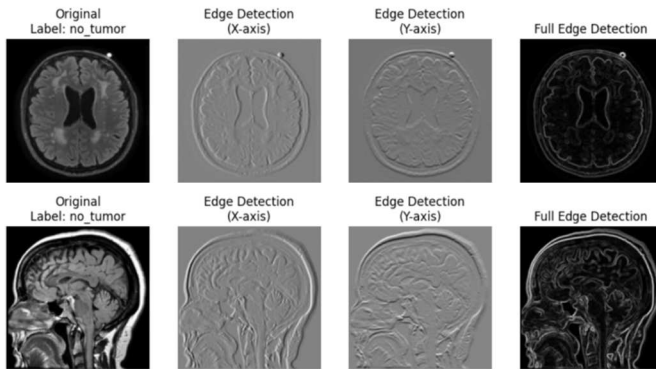


Fig.C1. Edge Detection using Scharr Operator

Accuracy: 0.68
Precision: 0.82
Recall: 0.68
F1-Score: 0.64
Matthews Correlation Coefficient: 0.6239
Cohen's Kappa: 0.5677

	precision	recall	f1-score	support
glioma_tumor	1.00	0.19	0.32	16
meningioma_tumor	0.88	0.95	0.91	11
no_tumor	0.50	1.00	0.66	16
pituitary_tumor	0.92	0.49	0.64	7
accuracy			0.68	39
macro avg	0.82	0.66	0.63	39
weighted avg	0.82	0.68	0.64	39

Fig.C2. Evaluated Values of the Model – III

Fig.C3. Confusion Matrix (III)

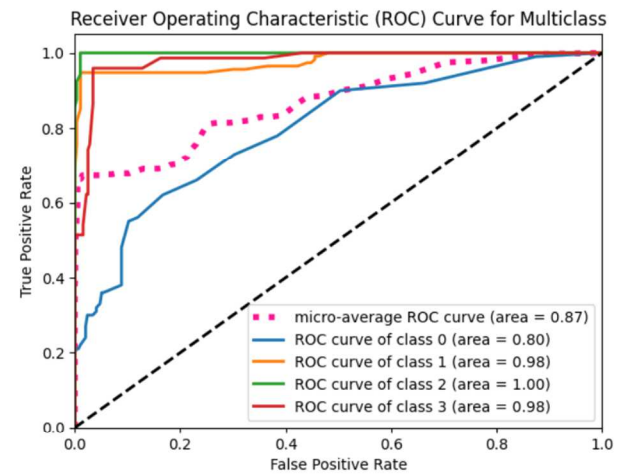


Fig.C4. AUC and ROC Curves (III)

Kirsch Mask

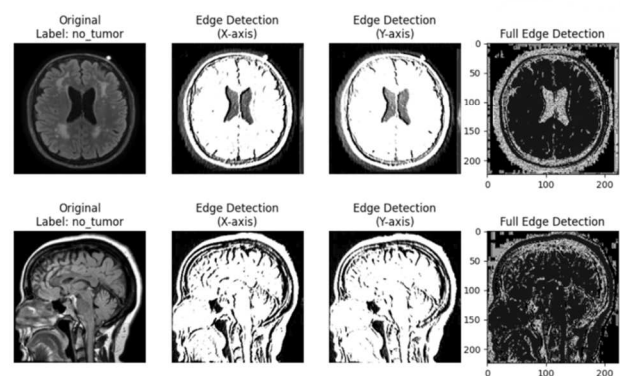


Fig.D1. Edge Detection using Kirsch Masking

Accuracy:	0.68			
Precision:	0.78			
Recall:	0.68			
F1-Score:	0.63			
Matthews Correlation Coefficient:	0.6072			
Cohen's Kappa:	0.5633			
	precision	recall	f1-score	support
glioma_tumor	1.00	0.19	0.32	100
meningioma_tumor	0.73	0.96	0.83	115
no_tumor	0.56	1.00	0.72	105
pituitary_tumor	0.87	0.46	0.60	74
accuracy			0.68	394
macro avg	0.79	0.65	0.62	394
weighted avg	0.78	0.68	0.63	394

Fig.D2. Evaluated Values of the Model – IV

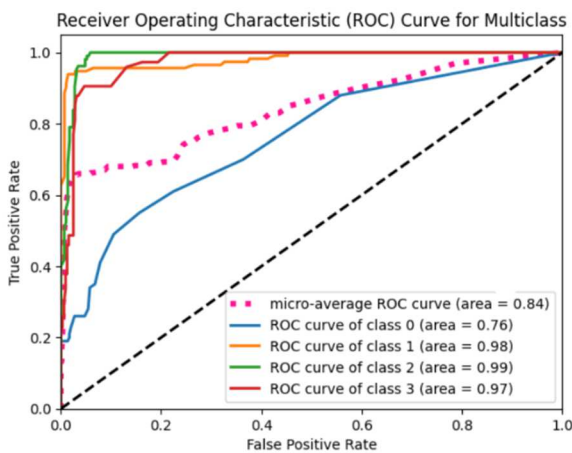


Fig.D3. AUC and ROC Curves (IV)

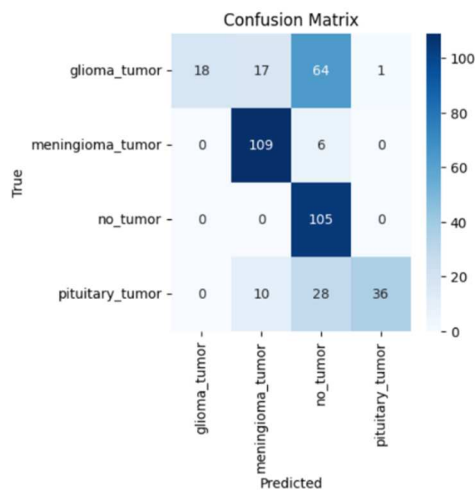


Fig.D4. Confusion Matrix (IV)

Frei – Chen Mask

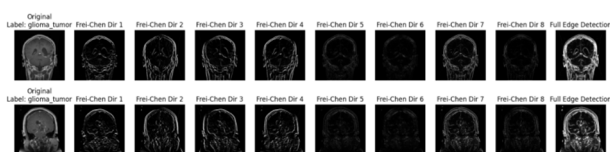


Fig.E1. Edge Detection using Frei - Chen Masking

Accuracy:	0.68			
Precision:	0.81			
Recall:	0.68			
F1-Score:	0.63			
Matthews Correlation Coefficient:	0.6157			
Cohen's Kappa:	0.5635			
	precision	recall	f1-score	support
glioma_tumor	1.00	0.18	0.31	100
meningioma_tumor	0.80	0.95	0.87	115
no_tumor	0.52	1.00	0.68	105
pituitary_tumor	0.97	0.49	0.65	74
accuracy			0.68	394
macro avg	0.82	0.65	0.63	394
weighted avg	0.81	0.68	0.63	394

Fig.E2. Evaluated Values of the Model – V

Fig.E3. AUC and ROC Curves (V)

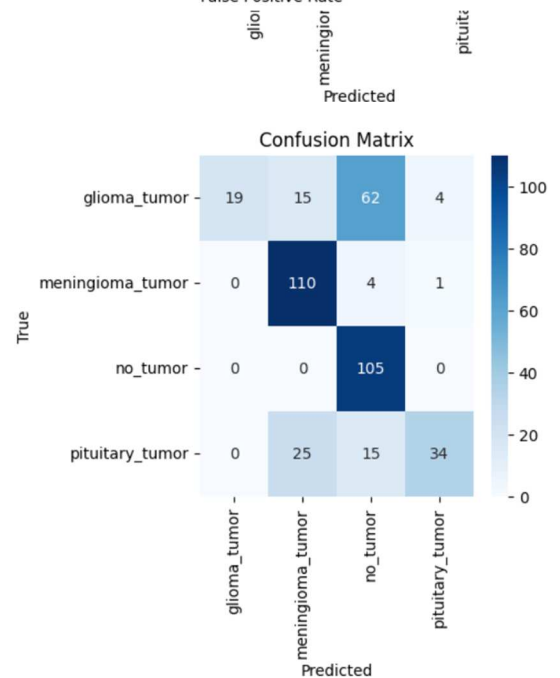
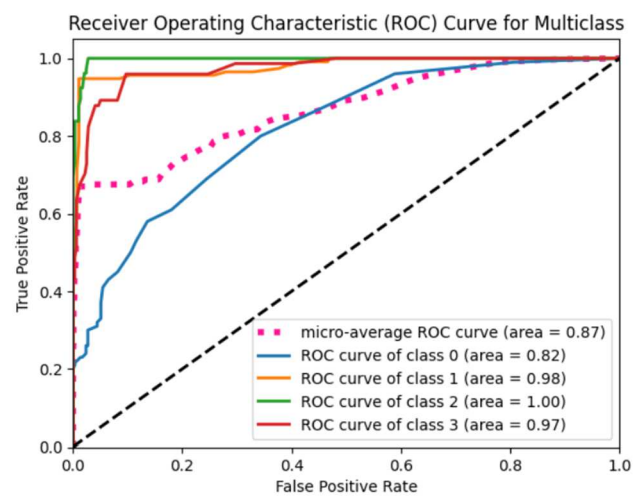


Fig.E4. Confusion Matrix (V)

Canny Operator

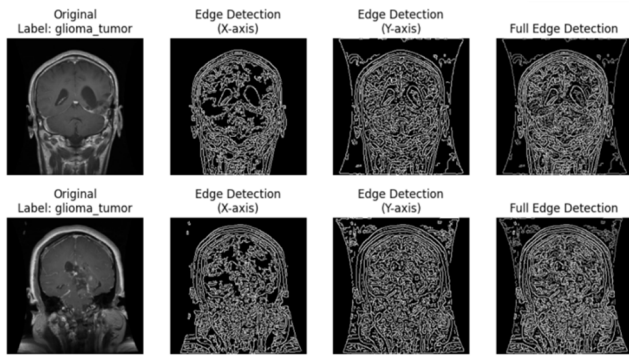


Fig.F1. Edge Detection using Canny Operator

Accuracy: 0.73
Precision: 0.76
Recall: 0.73
F1-Score: 0.68
Matthews Correlation Coefficient: 0.6538
Cohen's Kappa: 0.6296

	precision	recall	f1-score	support
glioma_tumor	0.92	0.23	0.37	100
meningioma_tumor	0.74	0.96	0.84	115
no_tumor	0.72	1.00	0.84	105
pituitary_tumor	0.63	0.65	0.64	74
accuracy			0.73	394
macro avg	0.75	0.71	0.67	394
weighted avg	0.76	0.73	0.68	394

Fig.F2. Evaluated Values of the Model – VI

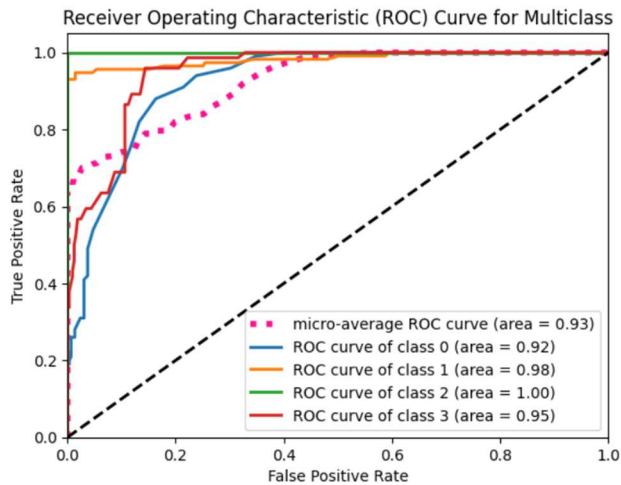


Fig.F3. AUC and ROC Curves (VI)

Confusion Matrix

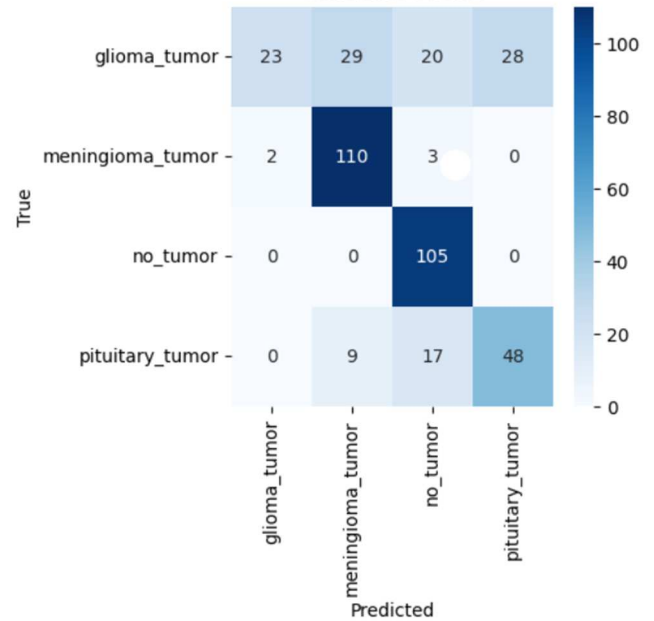


Fig.F4. Confusion Matrix (VI)

Laplacian of Gaussian (LoG)

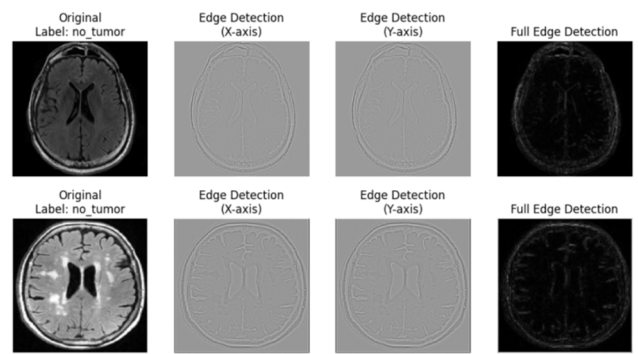


Fig.G1. Edge Detection using LoG Operator

Accuracy: 0.69
Precision: 0.80
Recall: 0.69
F1-Score: 0.65
Matthews Correlation Coefficient: 0.6309
Cohen's Kappa: 0.5815

	precision	recall	f1-score	support
glioma_tumor	0.95	0.21	0.34	100
meningioma_tumor	0.85	0.96	0.90	115
no_tumor	0.52	1.00	0.68	105
pituitary_tumor	0.93	0.50	0.65	74
accuracy			0.69	394
macro avg	0.81	0.67	0.64	394
weighted avg	0.80	0.69	0.65	394

Fig.G2. Evaluated Values of the Model – VII

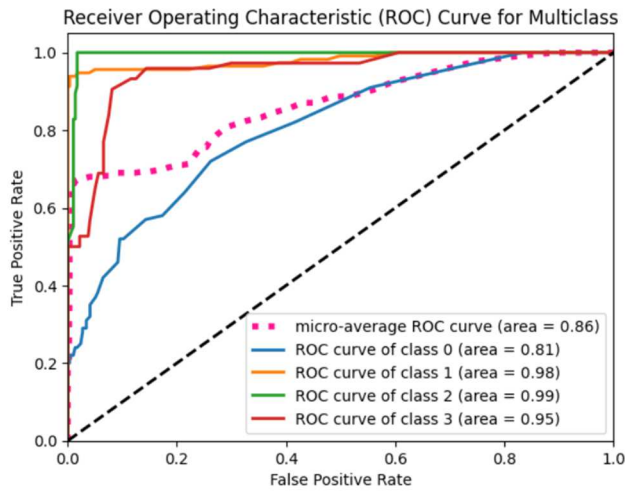


Fig.G3. AUC and ROC Curves (VII)

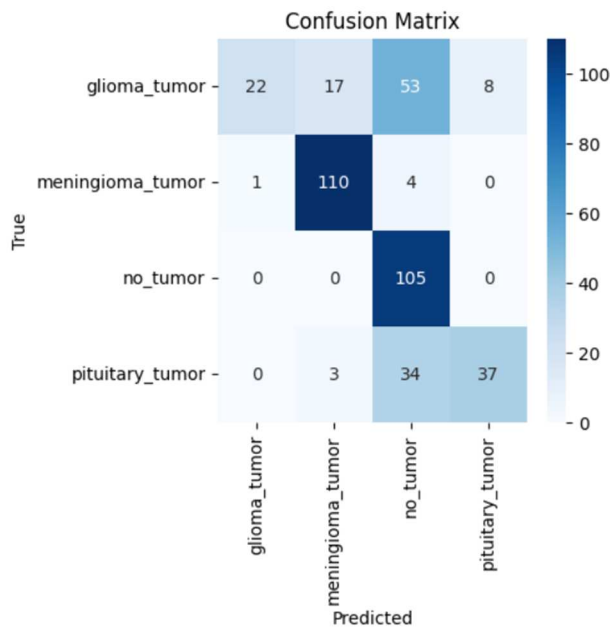


Fig.G4. Confusion Matrix (VII)

VI. CONCLUSION

Terminating the experiment directed, Sobel operator is seen to perform with accuracy of 86% and a precision factor of 80%. Canny operator stands second in the hierarchy, with an accuracy of 75% and precision of 79%. Roberts Zero-crossing and Laplacian of Gaussian operators have similar accuracy of 70% and a precision of 79% and 80%, respectively. Kirsch and Frei-Chen masks have comparatively the least accuracy of 68%, precision of 78%, and 69% accuracy and 80% precision, respectively.

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